

Section 9.1-9.2

n. 2

$$1. \text{eff}(\hat{\theta}_1, \hat{\theta}_5) = \frac{V(\hat{\theta}_5)}{V(\hat{\theta}_1)} = \frac{\sigma^2/n}{\sigma^2} = \frac{1}{n} < 1 \quad \text{when } n > 1$$

$$\text{eff}(\hat{\theta}_2, \hat{\theta}_5) = \frac{V(\hat{\theta}_5)}{V(\hat{\theta}_2)} = \frac{\sigma^2/n}{\frac{1}{4}(\sigma^2 + \sigma^2)} = \frac{\sigma^2/n}{\frac{1}{2}\sigma^2} = \frac{2}{n} < 1 \quad \text{when } n > 2$$

$$\text{eff}(\hat{\theta}_3, \hat{\theta}_5) = \frac{V(\hat{\theta}_5)}{V(\hat{\theta}_3)} = \frac{\sigma^2/n}{\frac{1}{9}(\sigma^2 + 4\sigma^2)} = \frac{\sigma^2/n}{\frac{5}{9}\sigma^2} = \frac{9}{5n} < 1 \quad \text{when } n > 5$$

Assuming $n=3$

$$\text{eff}(\hat{\theta}_1, \hat{\theta}_5) = \frac{1}{3}; \quad \text{eff}(\hat{\theta}_2, \hat{\theta}_5) = \frac{2}{3}; \quad \text{eff}(\hat{\theta}_3, \hat{\theta}_5) = \frac{3}{5}$$

$$2. E(\hat{\mu}) = E\left(\frac{1}{2}(Y_1 + Y_2)\right) = \frac{1}{2}(\mu + \mu) = \mu$$

$$E(\hat{\mu}_2) = E\left(\frac{1}{4}Y_1 + \frac{Y_2 + \dots + Y_{n-1}}{2(n-2)} + \frac{1}{4}Y_n\right) = \frac{1}{4}\mu + \frac{n(n-2)}{2(n-2)} \cdot \frac{1}{2}\mu + \frac{1}{4}\mu = \frac{1}{4}\mu + \frac{1}{2}\mu + \frac{1}{4}\mu = \mu$$

$$E(\hat{\mu}_3) = E(\bar{Y}) = \frac{1}{n} \sum \mu = \mu$$

Since $E(\hat{\mu}) = E(\hat{\mu}_2) = E(\hat{\mu}_3) = \mu$; These are all unbiased

$$b. \text{eff}(\hat{\mu}_3, \hat{\mu}_2) = \frac{V(\hat{\mu}_2)}{V(\hat{\mu}_3)} = \frac{\left(\frac{1}{16}\sigma^2 + \frac{1}{4(n-2)^2} \cdot (n-2)\sigma^2 + \frac{1}{16}\sigma^2\right)}{\frac{\sigma^2}{n}} = \left(\frac{1}{8}\sigma^2 + \frac{\sigma^2}{4(n-2)}\right) \times \frac{n}{\sigma^2} = n\left(\frac{1}{8} + \frac{1}{4(n-2)}\right)$$

$$\therefore \text{eff}(\hat{\mu}_3, \hat{\mu}_1) = \frac{V(\hat{\mu}_1)}{V(\hat{\mu}_3)} = \frac{\frac{\sigma^2}{4}}{\frac{\sigma^2}{n}} = \frac{n}{4}$$

→ See next pg. for a complete walkthrough

$$5. \text{eff}(\hat{\tau}_1^2, \hat{\tau}_2^2) = \frac{V(\hat{\tau}_2^2)}{V(\hat{\tau}_1^2)} = \frac{V\left(\frac{1}{2}(Y_1^2 - 2Y_1Y_2 + Y_2^2)\right)}{V(S^2)}$$

Rewrite S^2 as $\frac{n}{n-1} \left[\frac{1}{n} \sum_{i=1}^n Y_i^2 - \bar{Y}^2 \right]$

$$\Rightarrow \frac{1}{4} [V(Y_1^2) + 4V(Y_1Y_2) + V(Y_2^2)] \div \frac{n^2}{(n-1)^2} \left[\frac{1}{n^2} \sum_{i=1}^n V(Y_i^2) + V(\bar{Y}^2) \right]$$

$$6. \text{eff}(\hat{\tau}_1, \hat{\tau}_2) = \frac{V(\hat{\tau}_2)}{V(\hat{\tau}_1)} = \frac{V(\bar{Y})}{V\left(\frac{1}{2}(Y_1 + Y_2)\right)} = \frac{\sigma^2/n}{\frac{1}{4} \cdot 2\sigma^2} = \frac{2}{n}$$

#5: From Thm. 7.3, we know $\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$.

$$\text{So, } V\left[\frac{(n-1)S^2}{\sigma^2}\right] = 2(n-1) \Rightarrow \frac{(n-1)^2}{\sigma^4} V(S^2) = 2(n-1)$$

$$\Rightarrow V(S^2) = \frac{2\sigma^4}{n-1} = V(\hat{\sigma}_1^2)$$

Note: $Y_1 - Y_2 \sim N(0, \sqrt{2\sigma^2})$ (linear combination of Normal RVs)

$$\text{Thus, } Z = \frac{(Y_1 - Y_2) - 0}{\sqrt{2\sigma^2}} \sim N(0, 1).$$

$$\text{By Fact 2 (Chapter 7), } Z^2 = \frac{(Y_1 - Y_2)^2}{2\sigma^2} \sim \chi_1^2.$$

$$\text{Then, } V(\hat{\sigma}_2^2) = V\left[\frac{1}{2}(Y_1 - Y_2)^2\right] = V\left[\sigma^2 \cdot \frac{(Y_1 - Y_2)^2}{2\sigma^2}\right] = \sigma^4 V(Z^2) = 2\sigma^4.$$

$$\text{Hence, } \text{eff}(\hat{\sigma}_1^2, \hat{\sigma}_2^2) = \frac{2\sigma^4}{(2\sigma^4/(n-1))} = n-1.$$

$$8. f(y) = \begin{cases} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(y-\mu)^2} & ; y \geq 0 \\ 0, & \text{o.w} \end{cases}$$

$$I(\mu) = n E \left[- \frac{\partial^2 \ln[f(Y)]}{\partial \mu^2} \right] \quad \checkmark$$

$$\begin{aligned} \ln[f(Y)] &= \ln \left(\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(Y-\mu)^2} \right) \\ &= \ln \left(\exp \left[-\frac{1}{2\sigma^2}(Y-\mu)^2 \right] \right) - \ln(\sigma\sqrt{2\pi}) \\ &= -\frac{1}{2\sigma^2}(Y-\mu)^2 - \ln(\sigma\sqrt{2\pi}) \quad \checkmark \end{aligned}$$

$$\frac{\partial}{\partial \mu} \ln[f(Y)] = \frac{1}{\sigma^2} \cdot (Y-\mu) \quad \checkmark$$

$$\frac{\partial}{\partial \mu} \left[\frac{\partial}{\partial \mu} \ln[f(Y)] \right] = -\frac{1}{\sigma^2} \quad \checkmark$$

$$E \left[-\frac{1}{\sigma^2} \right] = -\frac{1}{\sigma^2}$$

$$I(\mu) = \frac{n}{\sigma^2} \quad \checkmark$$

$$\frac{1}{E(I)} = \frac{\sigma^2}{n} \quad \checkmark$$

$$E(\bar{Y}) = \frac{1}{n} \cdot n \mu = \mu \quad \checkmark$$

Thus $\bar{Y}$ is unbiased estimator for μ

$$V(\bar{Y}) = \frac{1}{n^2} \cdot n \sigma^2 = \frac{\sigma^2}{n} = \frac{1}{E(I)} \quad \checkmark$$

$\therefore \bar{Y}$ is efficient for μ $\checkmark$

* part (b.) missing $\left(-\frac{1}{2}\right)$

See next pg. for a complete walkthrough.

a.) First, $E(\bar{Y}) = \mu$, so $\bar{Y}$ is unbiased for μ .
 $V(\bar{Y}) = \frac{\sigma^2}{n}$

$$f(y) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(y-\mu)^2}{2\sigma^2}}$$

$$\ln f(y) = -\ln(\sigma\sqrt{2\pi}) - \frac{(y-\mu)^2}{2\sigma^2}$$

$$\frac{\partial \ln f(y)}{\partial \mu} = \frac{(y-\mu)}{\sigma^2}$$

$$\frac{\partial^2 \ln f(y)}{\partial \mu^2} = -\frac{1}{\sigma^2}$$

$$\therefore I(\mu) = [nE\left(\frac{1}{\sigma^2}\right)]^{-1} = \left(\frac{n}{\sigma^2}\right)^{-1} = \frac{\sigma^2}{n}$$

Since $V(\bar{Y}) = \frac{1}{I(\mu)}$, $\bar{Y}$ is an efficient estimator of μ .

b.) First, $E(\bar{Y}) = \lambda$, so $\bar{Y}$ is unbiased for λ .
 $V(\bar{Y}) = \frac{\lambda}{n}$

$$p(y) = \frac{\lambda^y e^{-\lambda}}{y!} \Rightarrow \ln p(y) = y \ln \lambda - \lambda - \ln(y!)$$

$$\frac{\partial \ln p(y)}{\partial \lambda} = \frac{y}{\lambda} - 1$$

$$\frac{\partial^2 \ln p(y)}{\partial \lambda^2} = -\frac{y}{\lambda^2}$$

$$\therefore I(\lambda) = [nE\left(\frac{y}{\lambda^2}\right)]^{-1} = \left(\frac{n\lambda}{\lambda^2}\right)^{-1} = \left(\frac{n}{\lambda}\right)^{-1} = \frac{\lambda}{n}$$

Since $V(\bar{Y}) = \frac{1}{I(\lambda)}$, $\bar{Y}$ is an efficient estimator of λ .